

Title:

Project 1: Building and Configuring a Linux Environment

Student Name:

Pelumi Johnson

Course:

Introduction to Linux

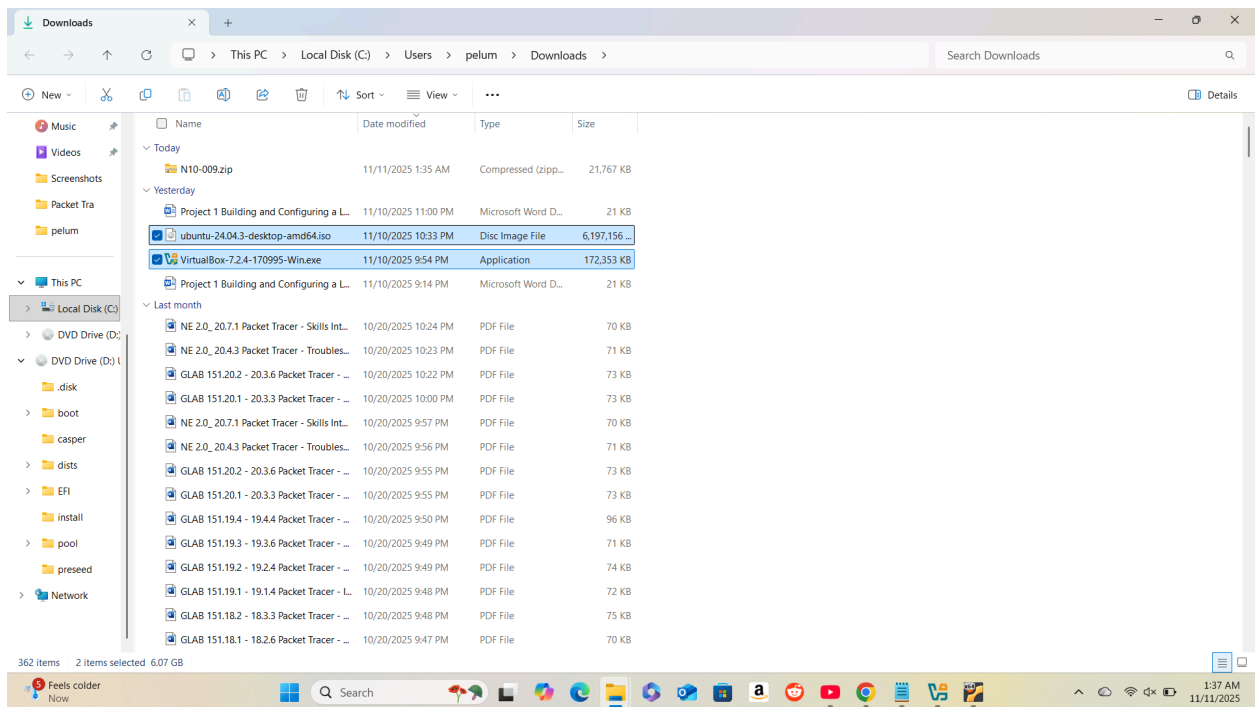
Submission Date:

11/11/2025

1.1 Part 1: Set Up a Virtual Machine

- Download and Install Virtual Machine Software

1.2 Download Linux ISO and Create a New Virtual Machine



Virtual machine name and operating system

VM Name

"Linux-Project1" 

VM Folder

 C:\Users\pelum\VirtualBox VMs 

ISO Image

<not selected>  

OS Edition



OS

Linux 

OS Distribution

Linux 2.6 

OS Version

Linux 2.6 / 3.x / 4.x / 5.x (64-bit) 

☐ Proceed with Unattended Installation



> Set up unattended guest OS installation

> Specify virtual hardware

> Specify virtual hard disk

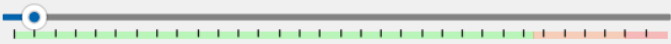
New Virtual Machine



Specify virtual hardware

Specify the VM's hardware. Resources allocated to the VM will not be available to the host when the VM is running.

Base Memory

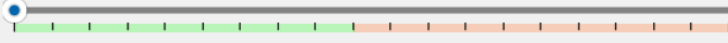


2000 MB 

4 MB

65536 MB

Number of CPUs



1 

1 CPU

20 CPUs

Disk Size



20.00 GB

4.00 MB

2.00 TB

☐ Use EFI

Help

Back

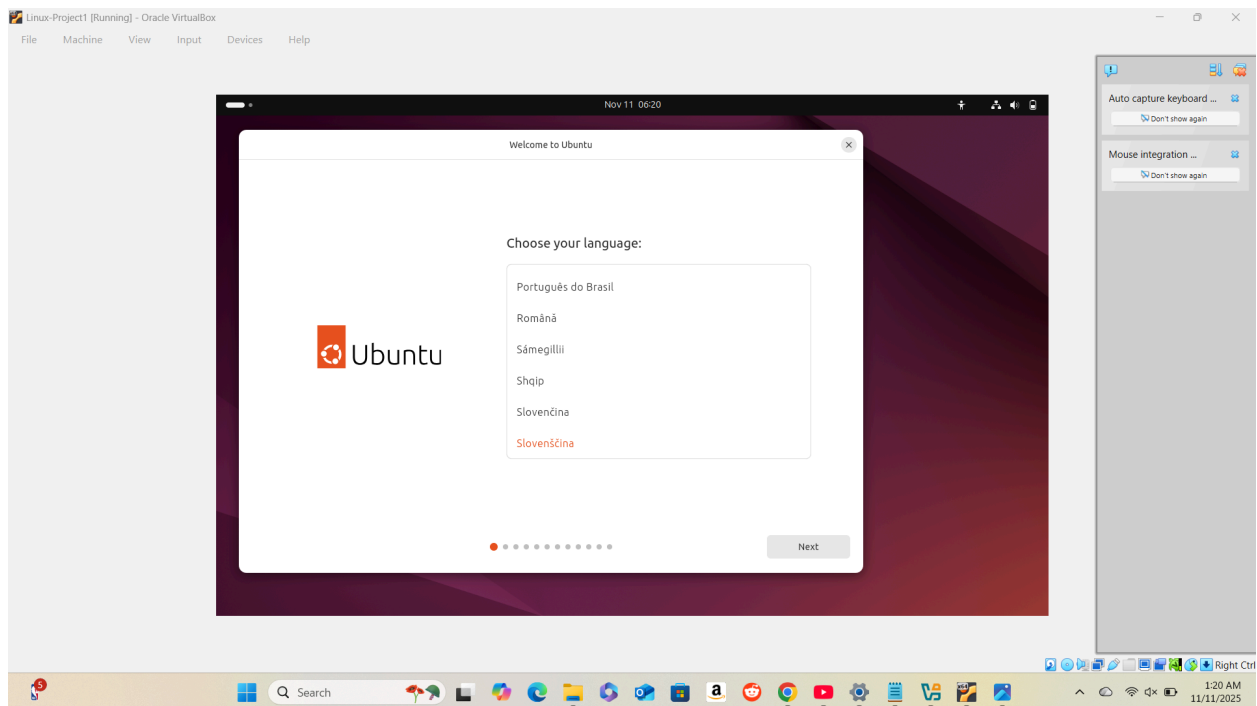
Next

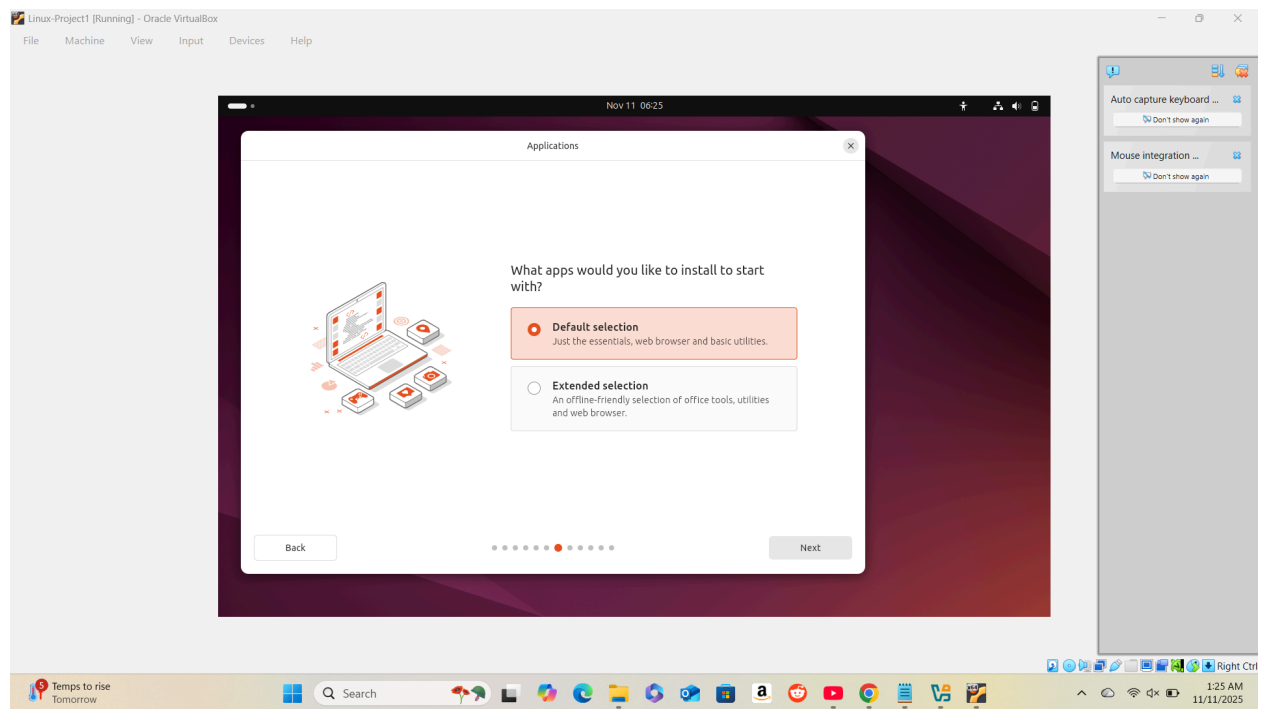
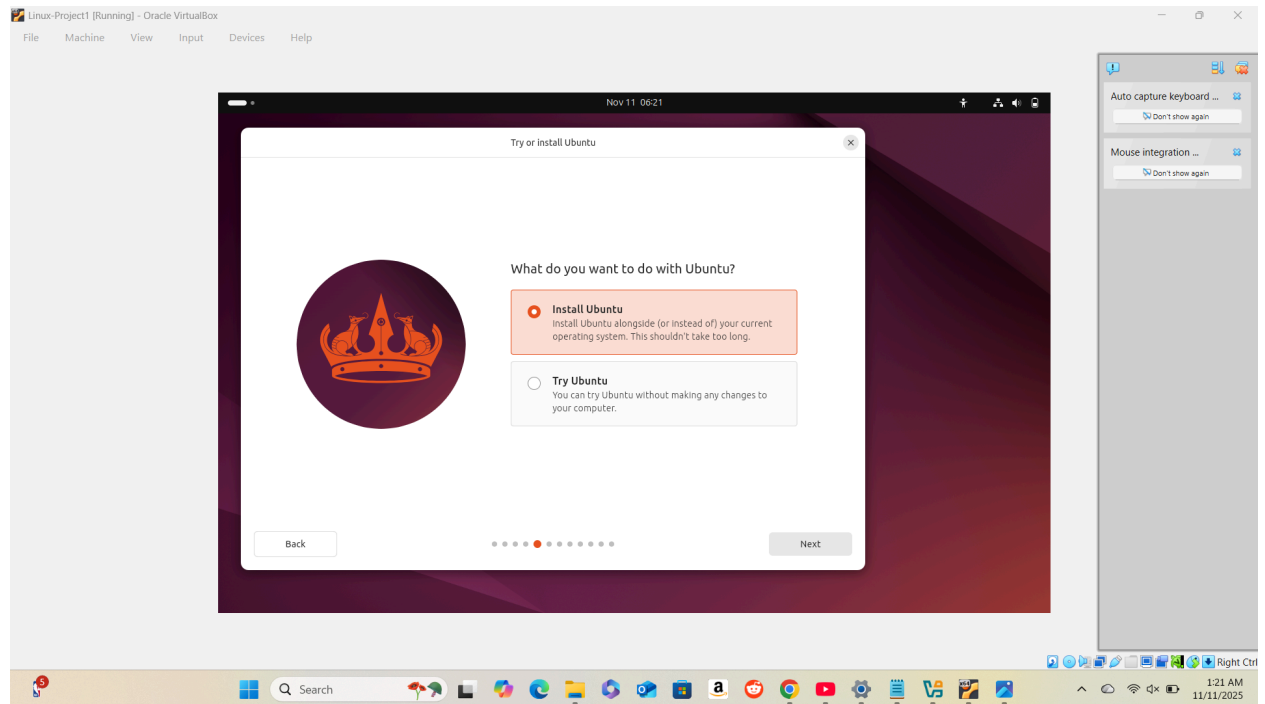
Cancel

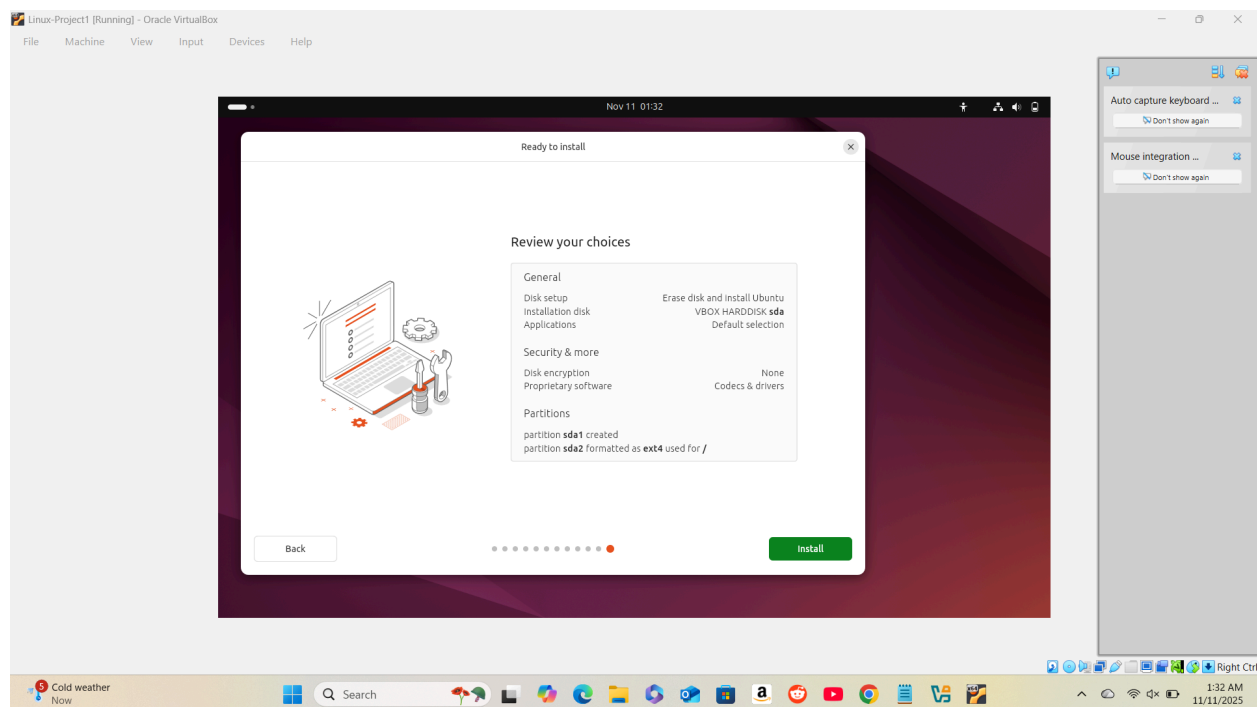
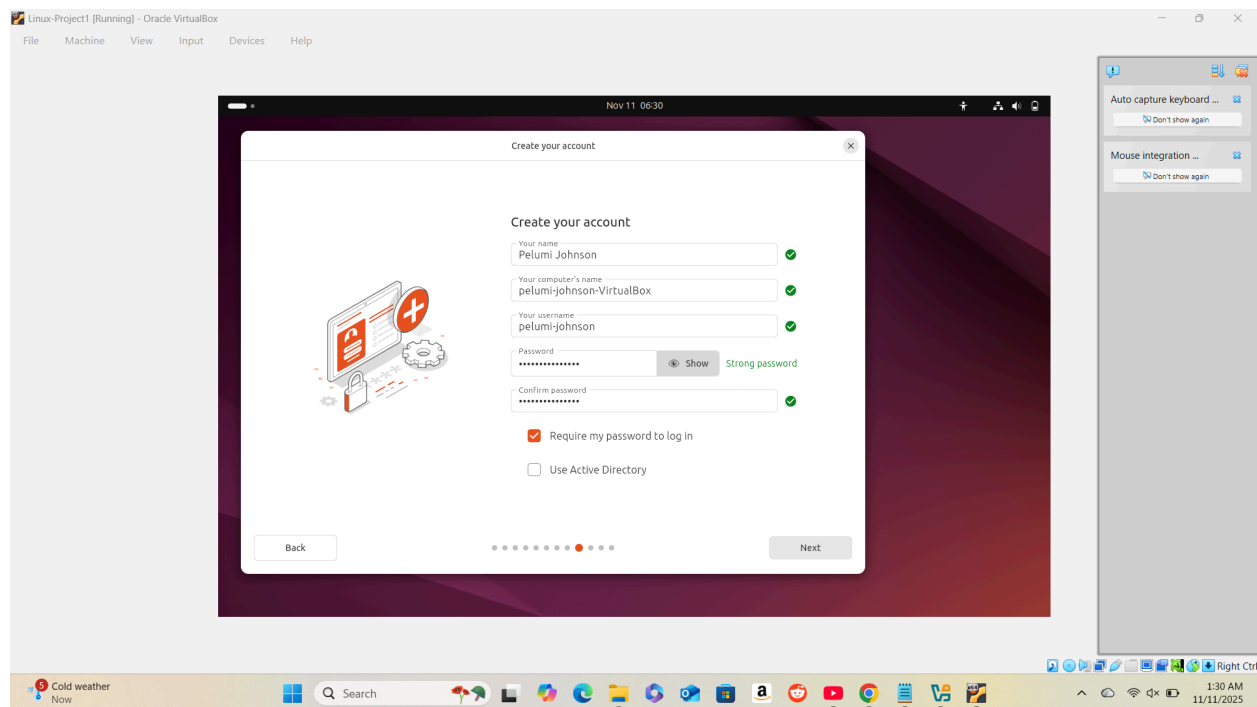
1.3 Install the Linux OS

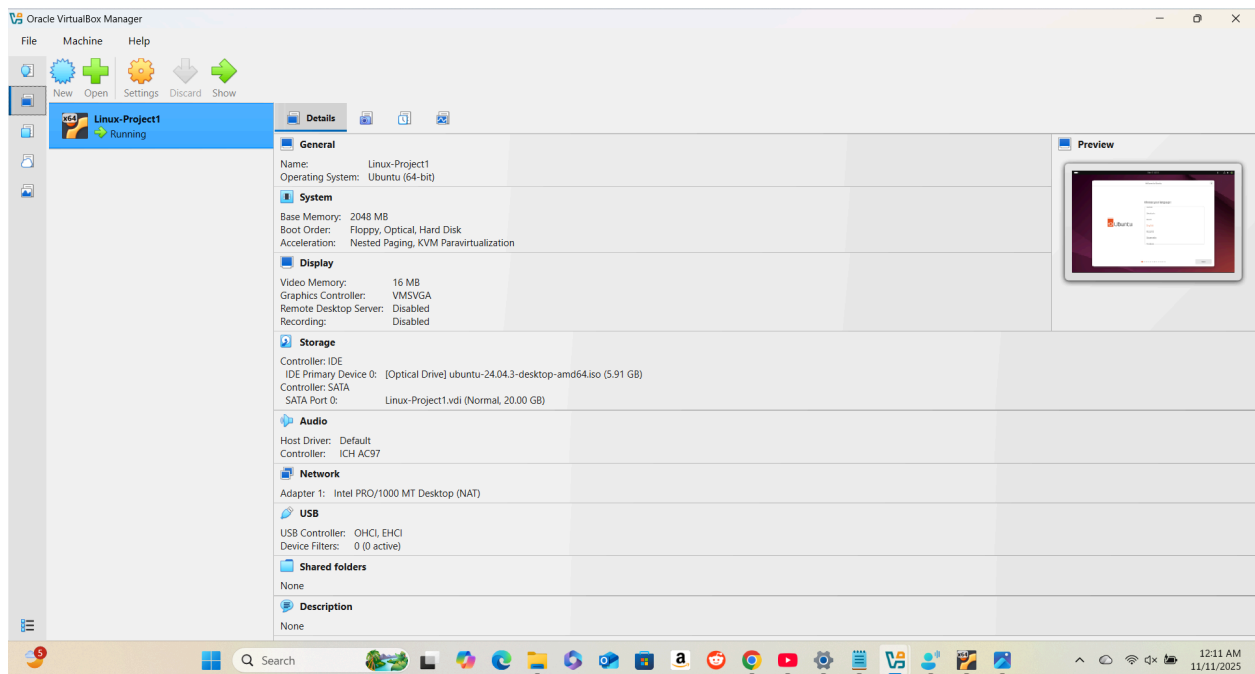
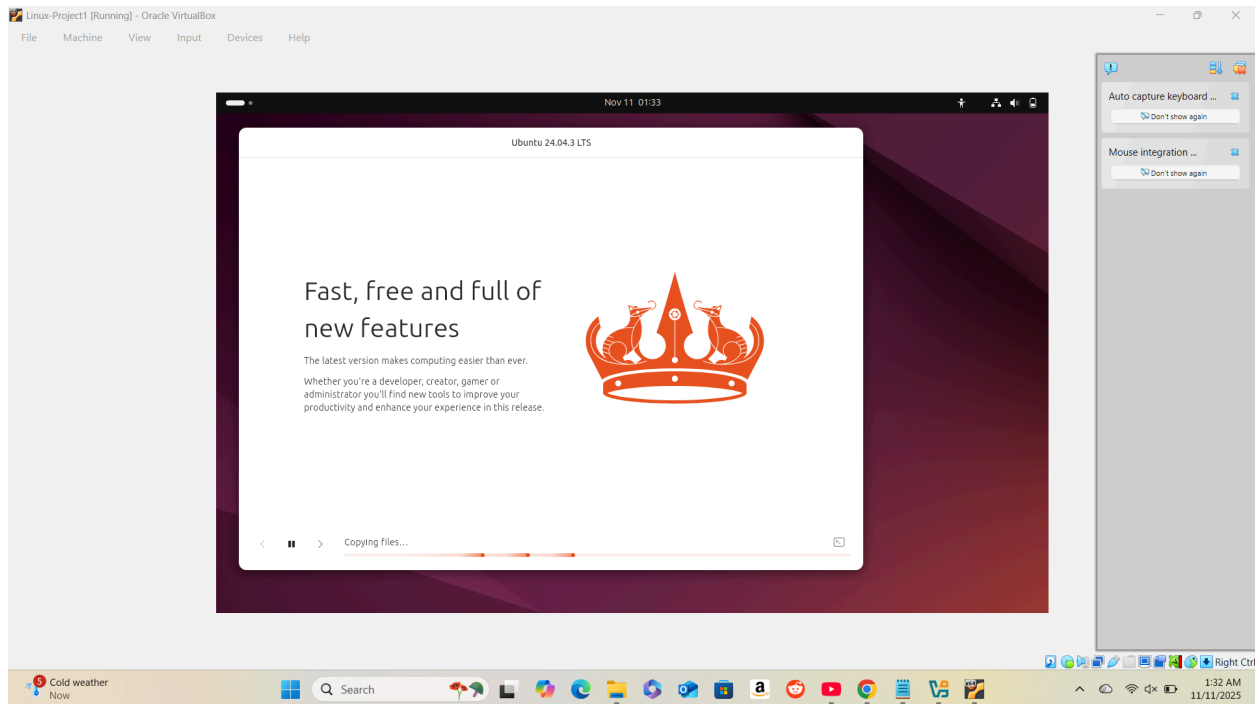
Describe the installation process and any custom settings used.

I installed Ubuntu 24.04.3 LTS using Oracle VirtualBox. First, I created a new virtual machine, allocated 2 GB of RAM and 25 GB of virtual hard drive space, and attached the Ubuntu ISO file. I selected the *interactive installation* option, chose English as the language, and used the default installation type (*Erase disk and install Ubuntu*). I enabled third-party software installation, selected the default app set, and created a user account named “Pelumi Johnson.” After installation, I restarted the VM and verified successful setup and internet connectivity using the `ping google.com` command in the Linux terminal.









1.4 Verify Installation

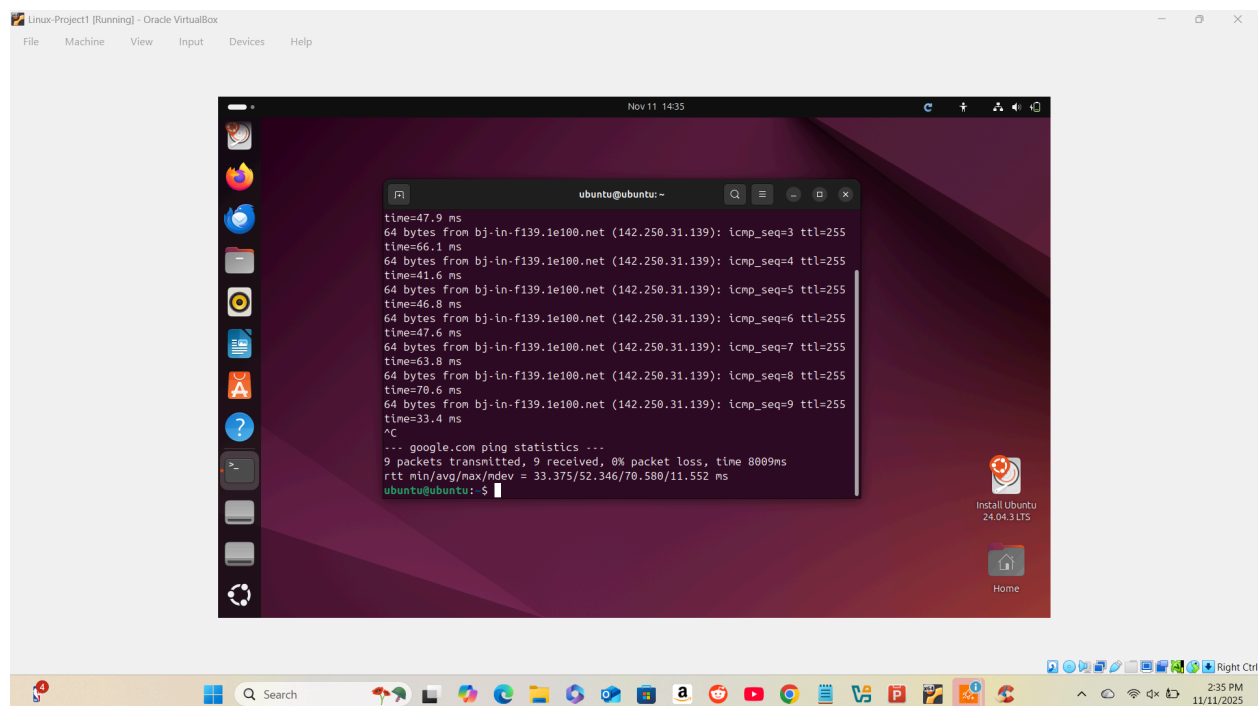
- *Record the command, its purpose, and the resulting output.*

After installation, I booted into the Ubuntu Linux environment. To confirm that the operating system and network were functioning correctly, I opened the Terminal and used the command:

ping google.com

Purpose: The ping command tests internet connectivity by sending data packets to a website and measuring the response time.

Result: The terminal displayed a series of replies such as “64 bytes from google.com...” , confirming that the system was successfully connected to the internet and functioning properly.



The screenshot shows a terminal window titled 'ubuntu@ubuntu:~' with the following output:

```
time=47.9 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=3 ttl=255
time=66.1 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=4 ttl=255
time=41.6 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=5 ttl=255
time=46.8 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=6 ttl=255
time=47.6 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=7 ttl=255
time=63.8 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=8 ttl=255
time=70.6 ms
64 bytes from bj-in-f139.1e100.net (142.250.31.139): icmp_seq=9 ttl=255
time=33.4 ms
^C
--- google.com ping statistics ---
9 packets transmitted, 9 received, 0% packet loss, time 8009ms
rtt min/avg/max/mdev = 33.375/52.346/70.580/11.552 ms
ubuntu@ubuntu:~$
```

2. Part 2: File System Exploration

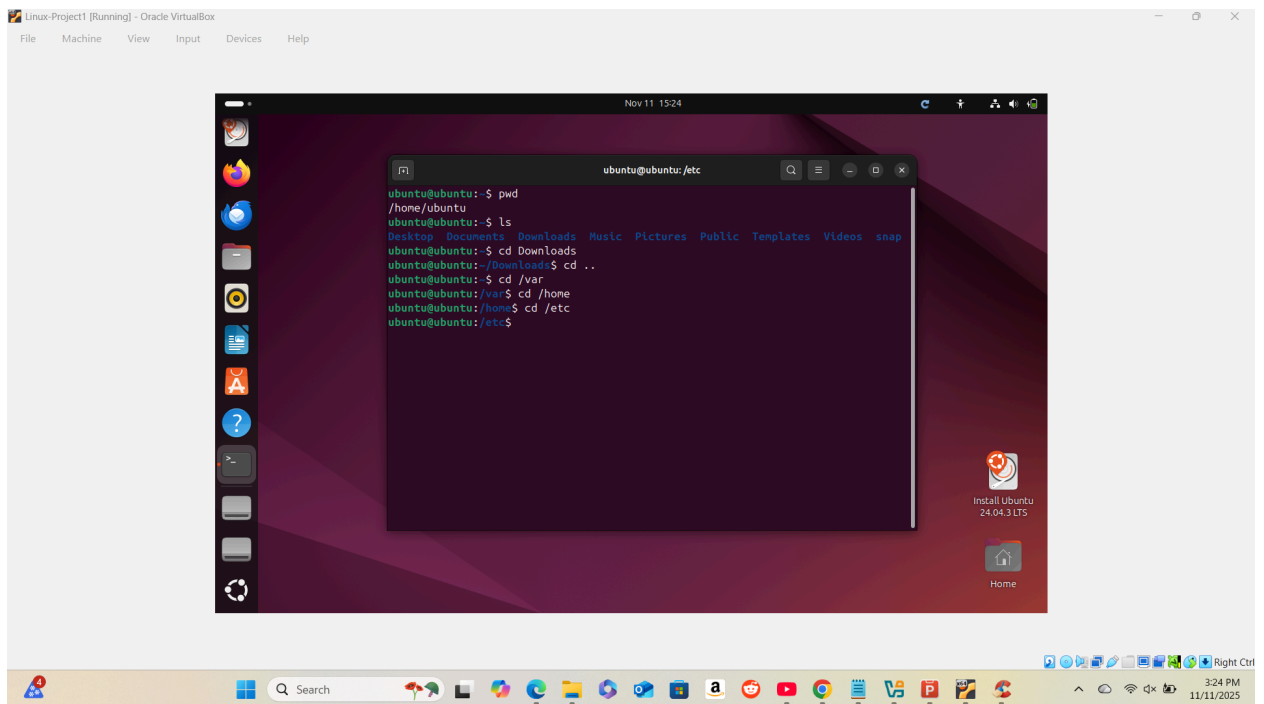
2.1 Navigating the File System

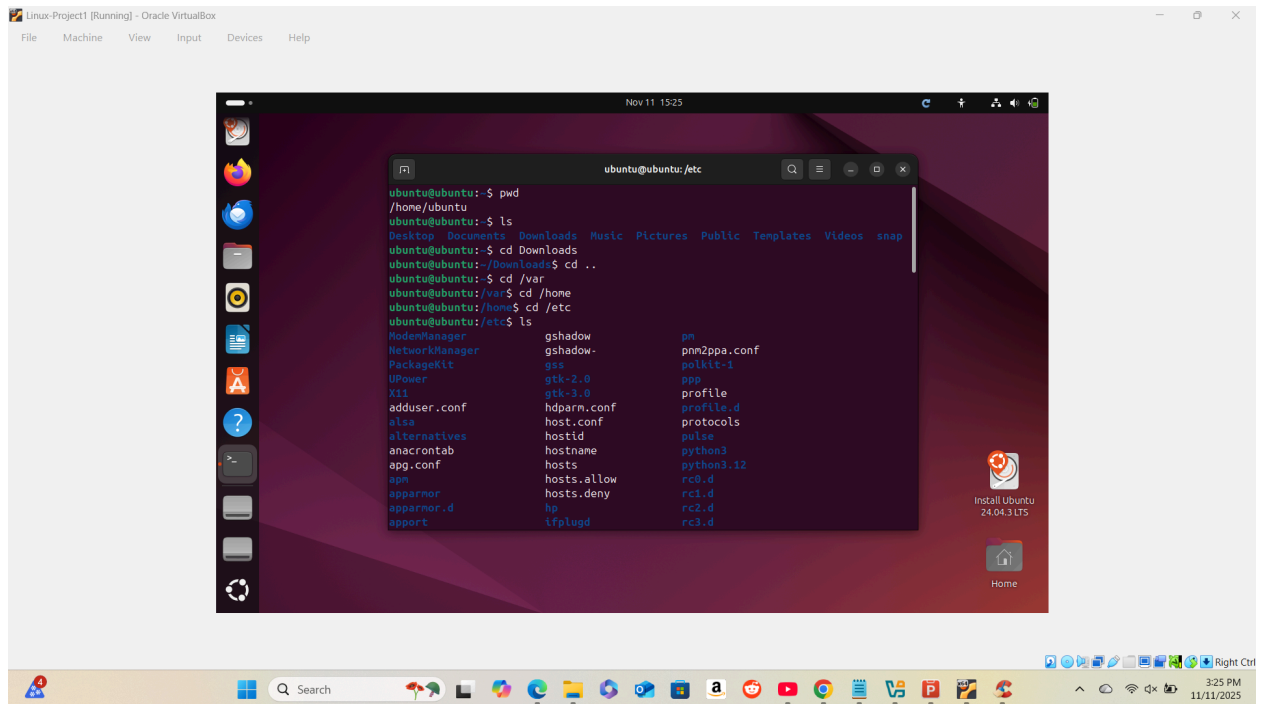
- Include a short explanation of what each command does and your observations.

- **pwd** – Prints the current working directory path.
- **ls** – Lists the files and folders in the current directory.
- **cd** – Changes from one directory to another.

Key directories explored:

- **/home** – Stores user files and personal folders.
- **/var** – Holds system logs and variable data.
- **/etc** – Contains system and software configuration files.





2.2 Create and Manipulate Files/Directories

- List each command with a brief description of its purpose.

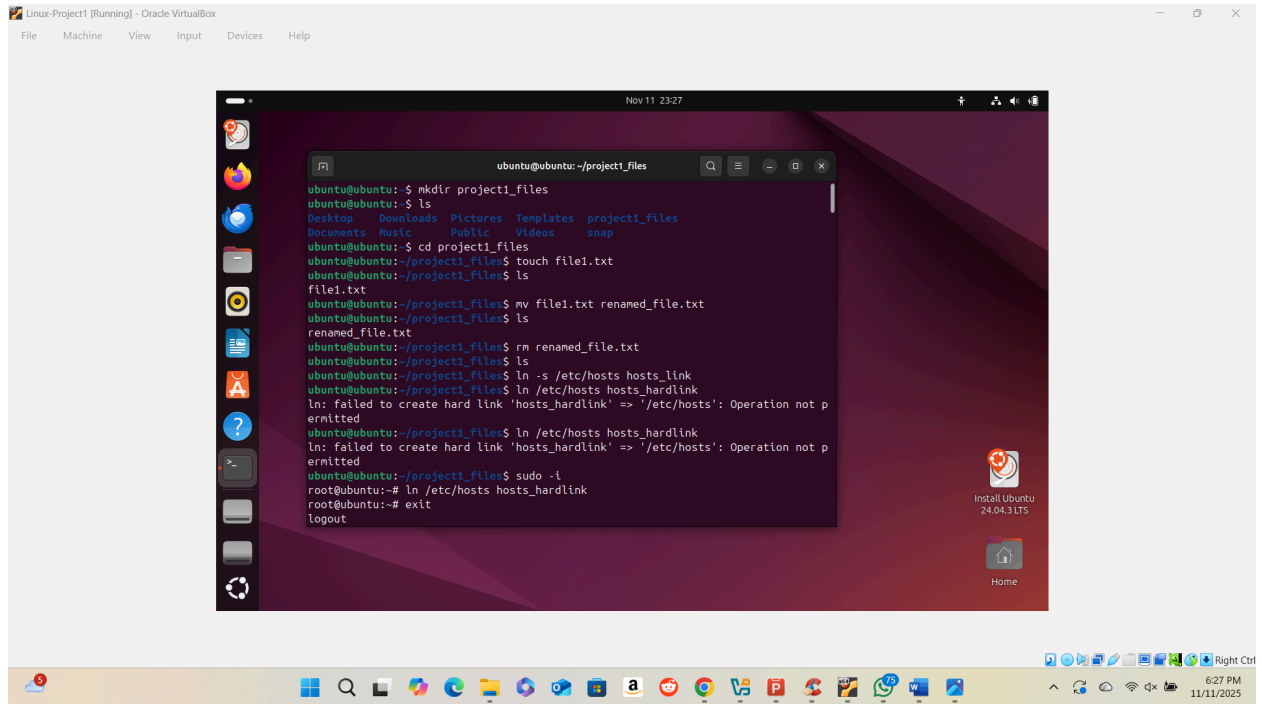
mkdir project1_files – Creates a new directory named project1_files.

cd project1_files – Moves into the newly created directory.

touch file1.txt – Creates a new, empty text file called file1.txt.

mv file1.txt renamed_file.txt – Renames file1.txt to renamed_file.txt.

rm renamed_file.txt – Deletes the file renamed_file.txt.



2.3 Create Links

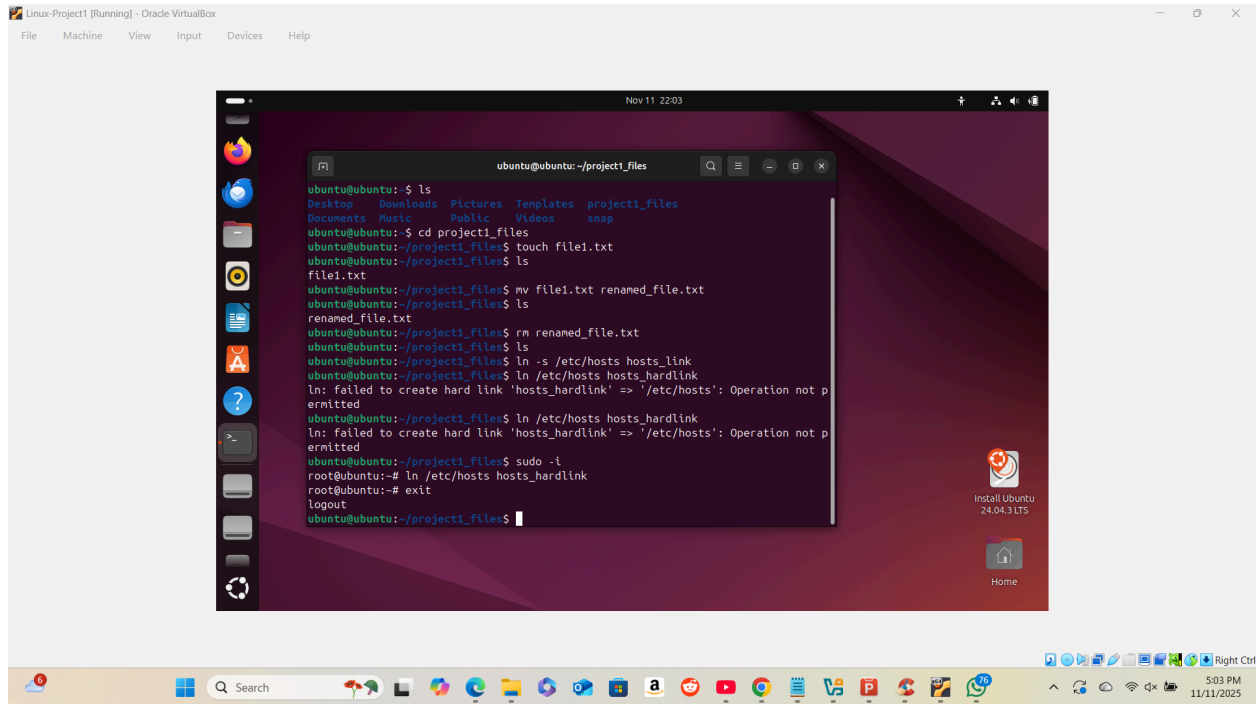
- Describe the difference between symbolic and hard links and document the commands.

Hard Link: Directly links to a file's data. It remains valid even if the original file is deleted.

Command: ln file1.txt file1_hardlink.txt

Symbolic Link: Acts as a shortcut to a file path. It breaks if the original file is deleted.

Command: ln -s file1.txt file1_symlink.txt



3.1 Create New Users and Groups

- Detail the commands used, their purpose, and any output observed.

sudo useradd student1 – Creates a new user named *student1*.

Output: No message if successful.

sudo passwd student1 – Sets a password for *student1*.

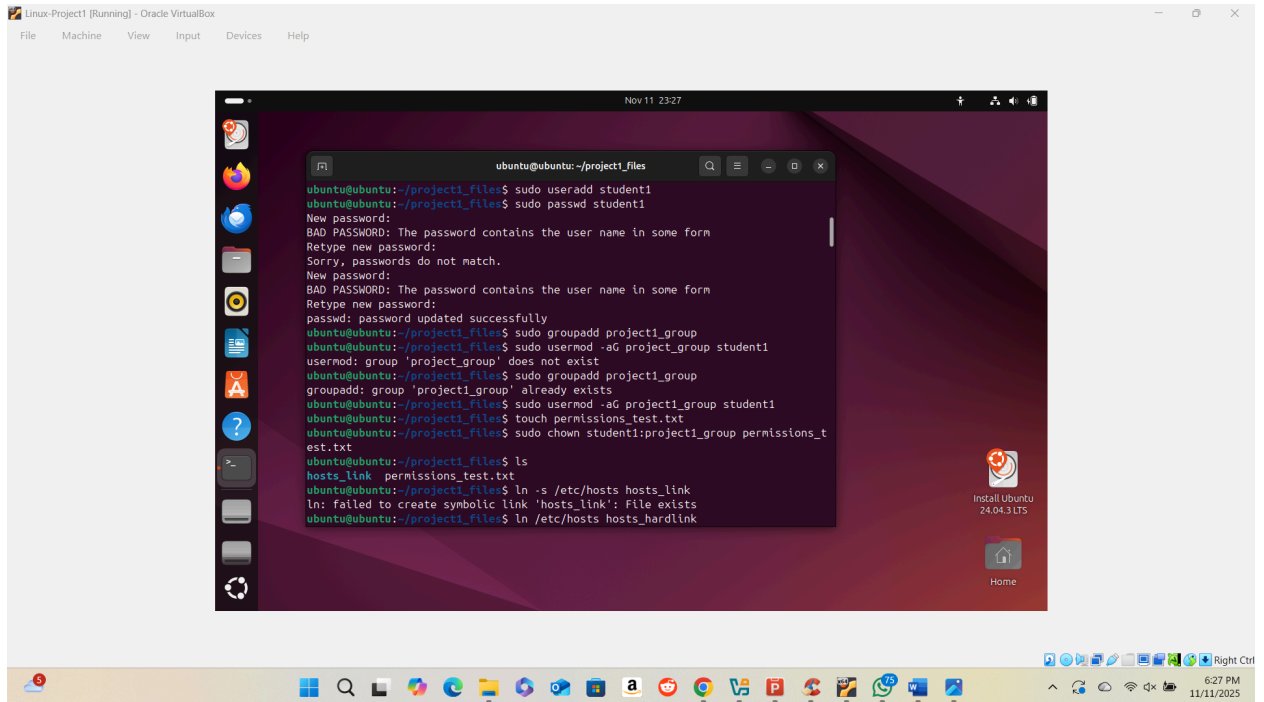
Output: Prompts to enter and confirm the new password.

sudo groupadd project1_group – Creates a new group named *project1_group*.

Output: No message if successful.

sudo usermod -aG project1_group student1 – Adds *student1* to the *project1_group* group.

Output: No message if successful.

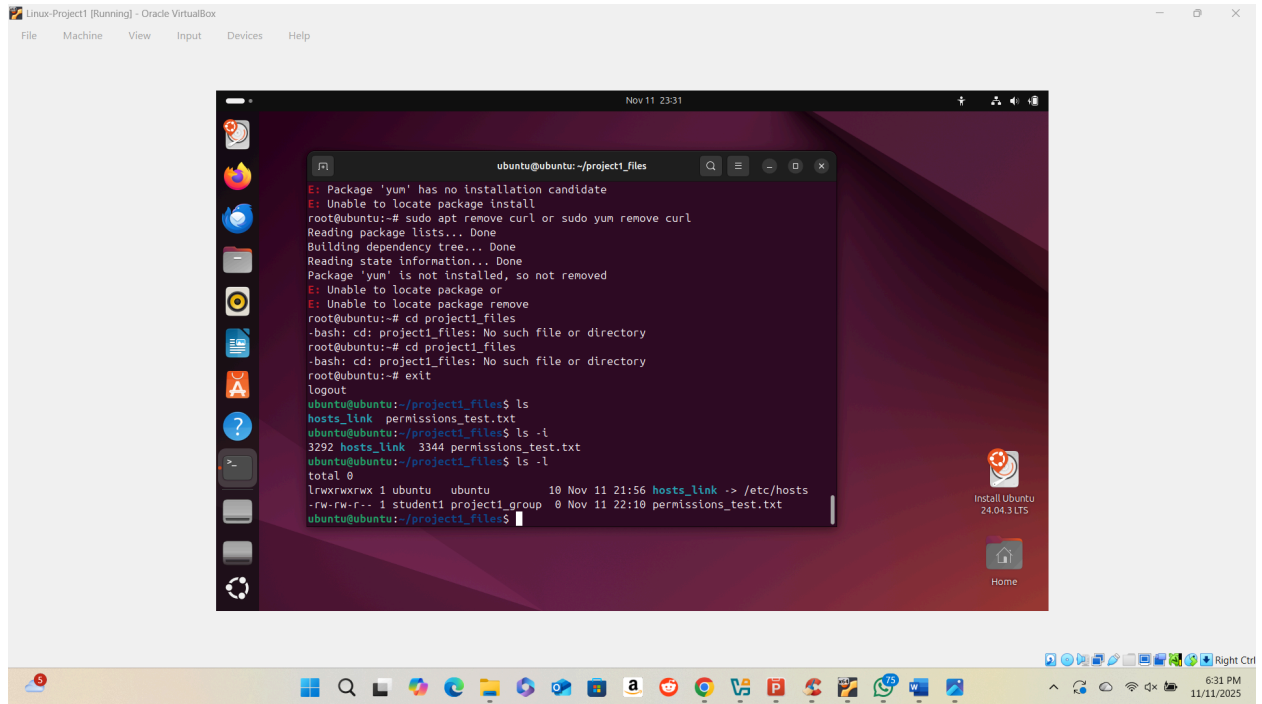


3.2 Modify File Permissions

- Explain what each permission command achieves.

chmod u+rw permissions_test.txt – Gives the file's owner (user) permission to **read and write** the file.

chmod o-w permissions_test.txt – Removes the **write** permission from “others,” preventing them from editing the file.



4. Part 4: Package Management

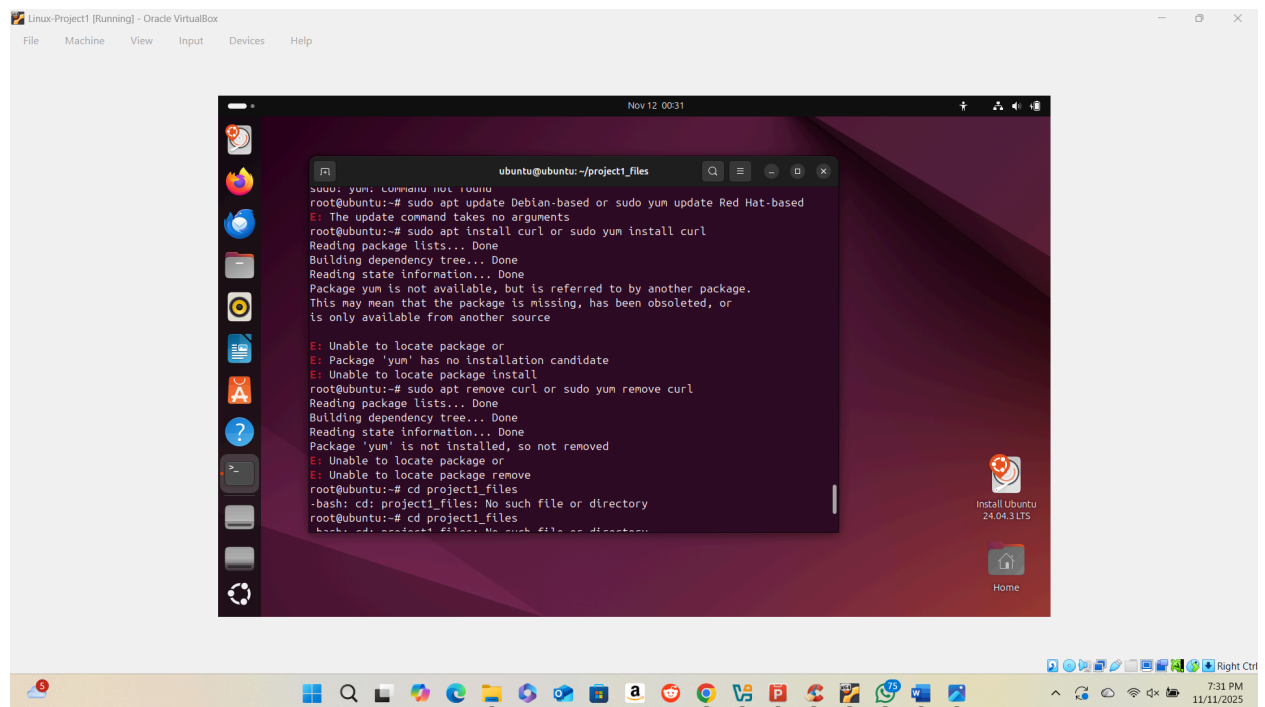
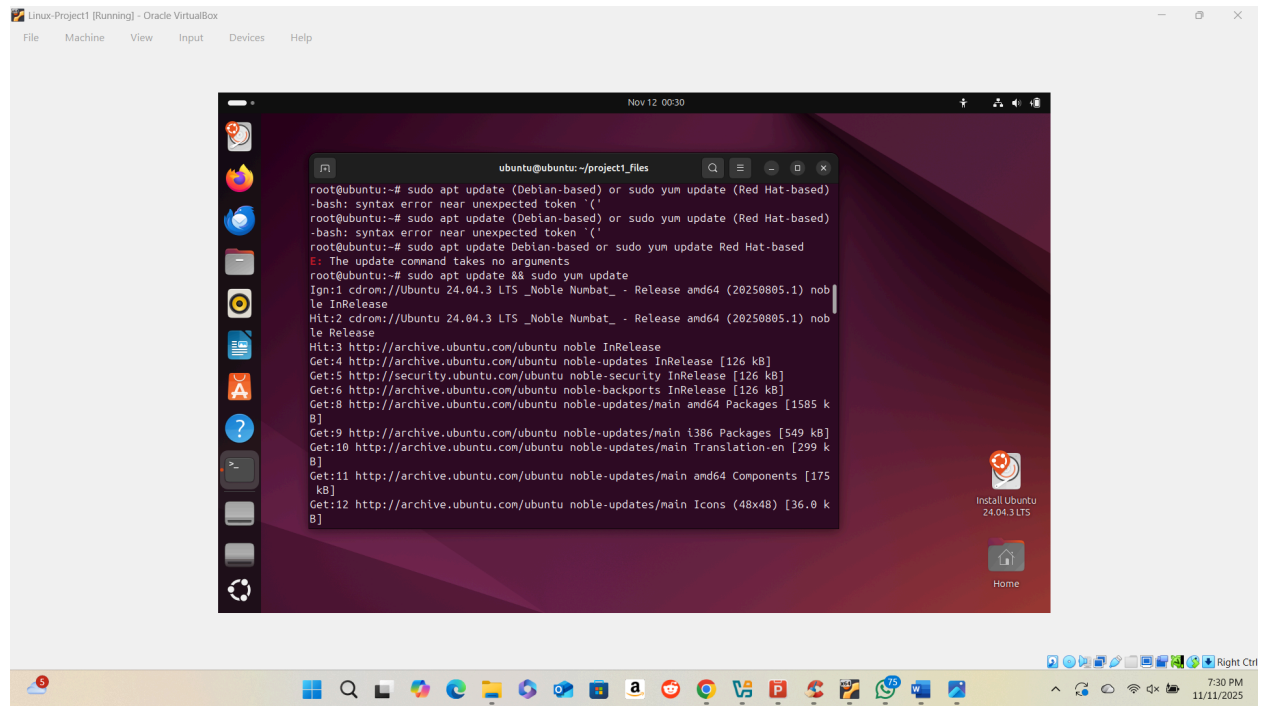
4.1 Install a Package

- List the commands with explanations and expected outcomes.

sudo apt update – Updates the list of available software packages (keeps the system up to date).

sudo apt install curl – Installs the curl package, a tool used to transfer data from or to a server.

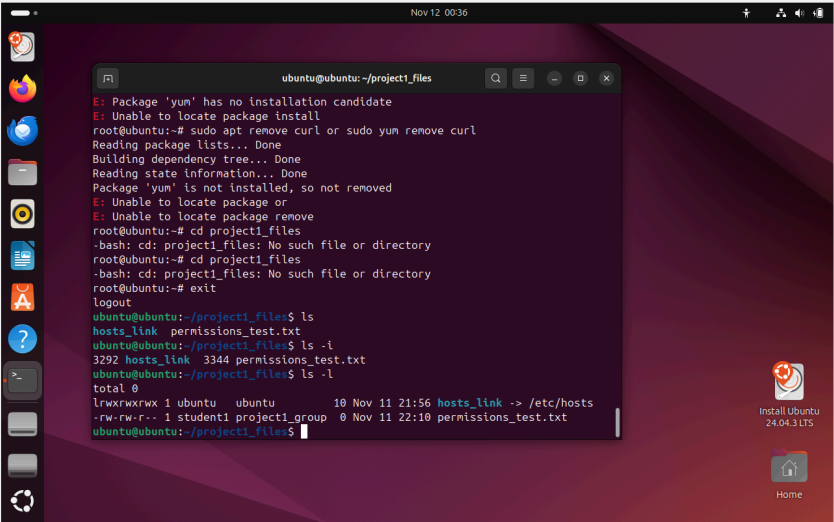
curl --version – Verifies the installation by displaying the installed version of curl.



4.2 Uninstall a Package

- Describe the removal process and verify with a screenshot.

sudo apt remove curl – Uninstalls the *curl* package from the system.



The screenshot shows a terminal window titled 'ubuntu@ubuntu:~/project1_files' with the following output:

```
ubuntu@ubuntu:~/project1_files$ sudo apt remove curl
E: Package 'yum' has no installation candidate
E: Unable to locate package install
root@ubuntu:~# sudo apt remove curl or sudo yum remove curl
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
Package 'yum' is not installed, so not removed
E: Unable to locate package or
E: Unable to locate package remove
root@ubuntu:~# cd project1_files
-bash: cd: project1_files: No such file or directory
root@ubuntu:~# cd project1_files
-bash: cd: project1_files: No such file or directory
root@ubuntu:~# exit
logout
ubuntu@ubuntu:~/project1_files$ ls
hosts_link  permissions_test.txt
ubuntu@ubuntu:~/project1_files$ ls -l
3292 hosts_link 3344 permissions_test.txt
ubuntu@ubuntu:~/project1_files$ ls -l
total 0
lrwxrwxrwx 1 ubuntu ubuntu 10 Nov 11 21:56 hosts_link -> /etc/hosts
-rw-rw-r-- 1 student1 project1_group 0 Nov 11 22:10 permissions_test.txt
ubuntu@ubuntu:~/project1_files$
```

5. Conclusion

Provide a summary of your experience completing the project. Include reflections on:

1. What I Learned:

I learned how to install and use Linux through VirtualBox, manage files and users, and use terminal commands confidently.

2. Challenges Encountered:

I faced setup and boot errors during installation, but I fixed them by restarting, recreating the virtual machine, and carefully following each step.

3. Future Considerations:

I plan to keep practicing Linux commands, explore system administration tasks, and strengthen my skills in managing and securing systems.

References

Ubuntu Documentation. (n.d.). Ubuntu tutorials and guides. Retrieved from <https://ubuntu.com/tutorials>

Oracle Corporation. (n.d.). VirtualBox User Manual. Retrieved from <https://www.virtualbox.org/manual/UserManual.html>